

Semester Test 3

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Electricity and Electronics EBN122

24 October 2025

Assessment Information

Maximum marks:	46	Full marks:	45
Duration of exam:	90 (excluding additional time)	Open/Closed book:	Closed
Additional time allocations:	5 min before and 10 min after for OCR answer sheet completion	Allowable materials:	Pencil case, non-programmable calculator
Extra-time venue:	Eng. 3, Netlab C	Submission format:	OCR
Total number of pages (this page included)		17	

Lecturers: Prof. JP de Villiers, Mr. S Twala, Prof. JP Jacobs**Important**

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of OCR answer sheet 00/01.
4. Read the instructions on OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. Final answers must be written as real numbers and not as fractions.
7. Use **at least 3 significant digits** to write real numbers. This means that the real and imaginary parts of complex numbers should each be written using at least 3 significant digits, for example $Z = 10.0 + j0.283 \Omega$ (the same applies to magnitudes and angles).
8. Exact numbers in which decimal expansions are zero, can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

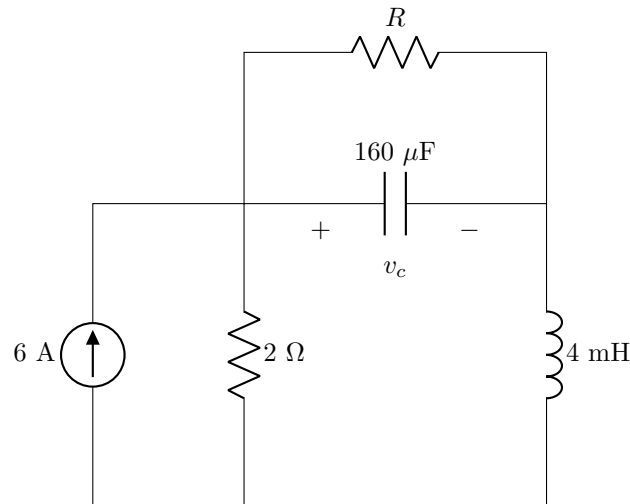
INSTRUCTIONS**Do step 0 in the first 5 minutes of the assessment. (-5 to 0 min)****Do steps 1 and 2 in the next 100 minutes of the assessment. (0-100 min)****TIME: -5 min to 0 min****STEP 0 :** Complete the front page of OCR answer sheet 01, and also fill in your student number on additional OCR answer sheets.**TIME: 0 min to 100 min****STEP 1:** Do your calculations on the question paper.**STEP 2:** Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN122S03**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

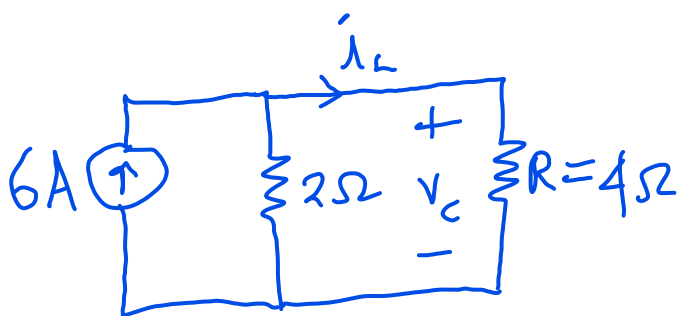
Questions 1 to 2 [3]

The circuit below operates under steady dc conditions.



- | | | |
|-----------|--|-----------|
| 01 | Find v_c if $R = 4 \Omega$. [1] | $8V$ |
| 02 | Find the value of R such that the energy stored in the capacitor is the same as that stored in the inductor. [2] | 5Ω |

① For DC:



Ohm's law:

$$v_c = \frac{R \times 2}{R + 2} \times 6$$

$$v_c = \frac{4 \times 2}{4 + 2} \times 6$$

$$v_c = 8V \rightarrow$$

② $w_c = \frac{1}{2} C v_c^2$
 $w_L = \frac{1}{2} L i_L^2$

$$w_c = w_L$$

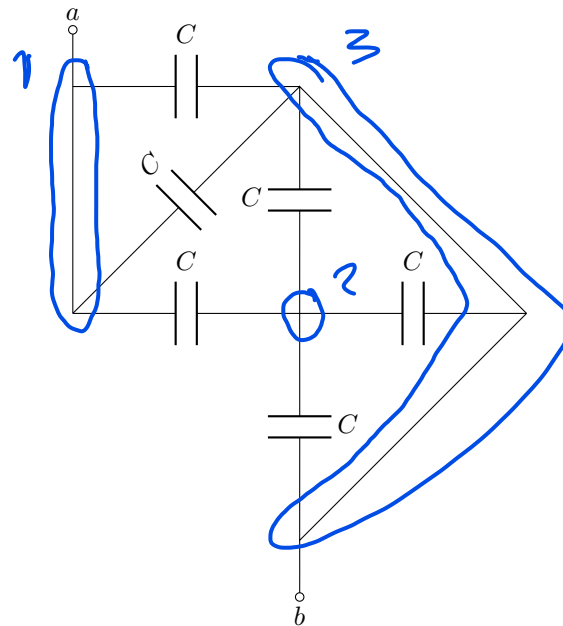
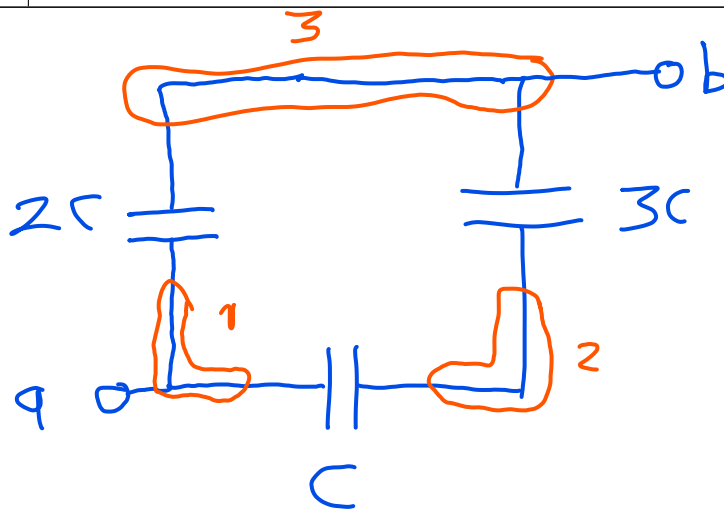
$$\frac{1}{2} C v_c^2 = \frac{1}{2} L i_L^2$$

$$C [i_L R]^2 = L i_L^2$$

$$R = \pm \sqrt{\frac{L}{C}}$$

$$R = \pm \sqrt{\frac{4 \times 10^{-3}}{160 \times 10^{-6}}}$$

$$R = 5\Omega \rightarrow$$

Question 3 [2]Given: $C = 20 \mu\text{F}$.**03**Determine the equivalent capacitance C_{ab} with respect to terminals a-b. [2] **$55 \mu\text{F}$** 

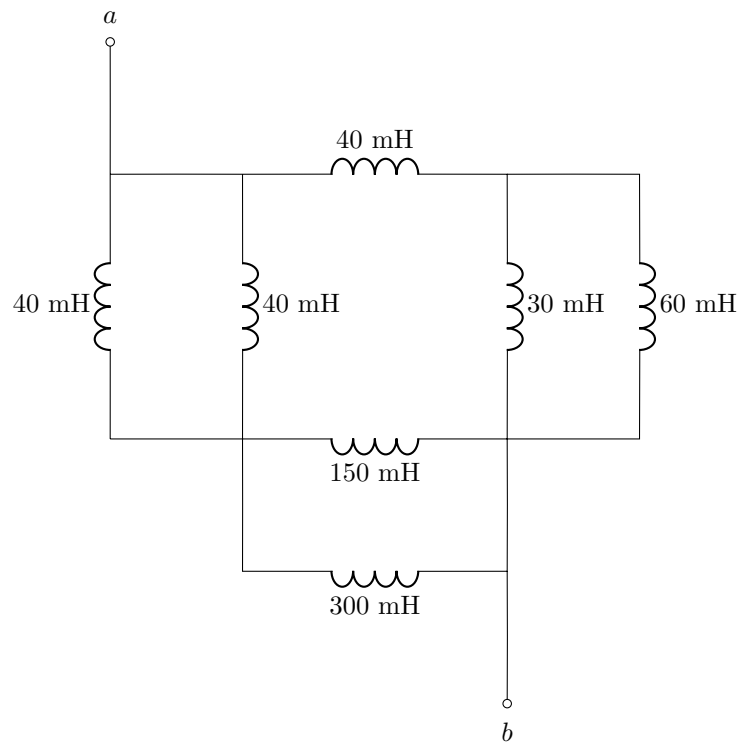
$$C_{ab} = 2C + \frac{(3C)(C)}{3C + C}$$

$$C_{ab} = 2C + \frac{3C^2}{4C}$$

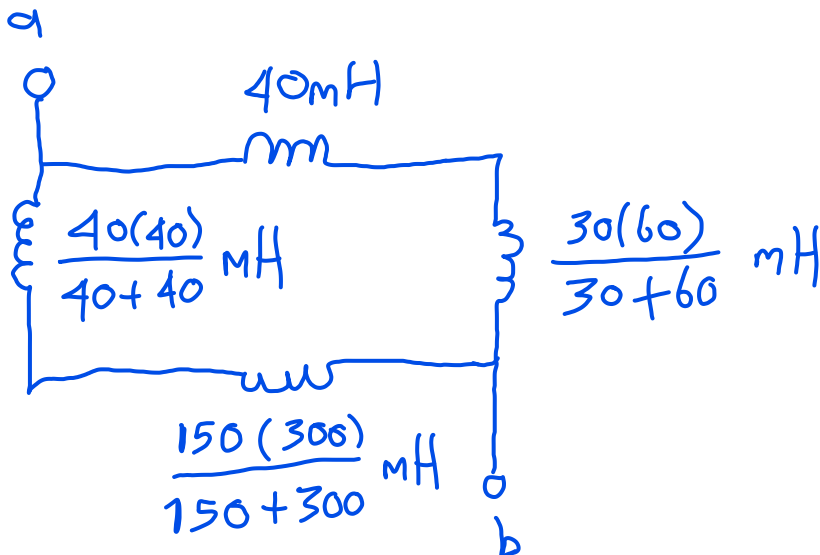
$$C_{ab} = 2(20 \times 10^{-6}) + \frac{3(20 \times 10^{-6})}{4}$$

$$C_{ab} = \underline{55 \mu\text{F}} \rightarrow$$

Question 4 [2]



04

Find the equivalent inductance L_{ab} when looking into the terminals $a - b$. [2]40 mH

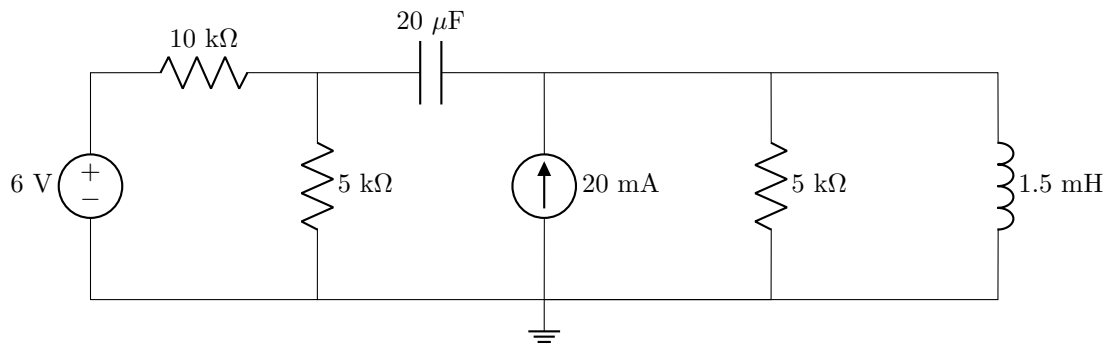
$$L_{ab} = \left[40 + \frac{30(60)}{30+60} \right] \parallel \left[\frac{40(40)}{40+40} + \frac{150(300)}{150+300} \right]$$

$$L_{ab} = \frac{60(120)}{60+120}$$

$$L_{ab} = \underline{40 \text{ mH}}$$

Questions 5 to 6 [4]

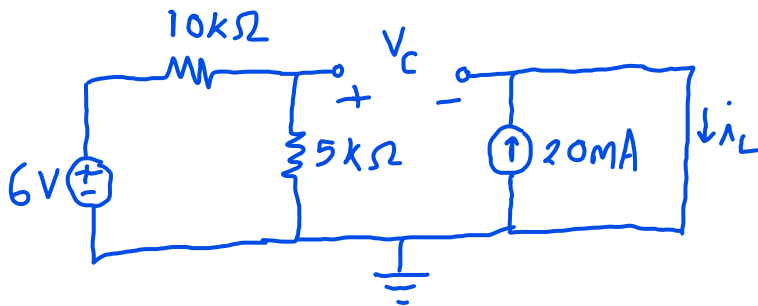
Under steady state dc conditions, analyse the circuit below.

05 Calculate the energy ω_L stored in the inductor L . [2]

300 nJ

06 Calculate the energy ω_C stored in the capacitor C . [2]

40 μJ



$$V_C = \frac{0.6}{5+10} \times 6$$

$$V_C = 2V$$

05

$$\omega_L = \frac{1}{2} L i_L^2$$

$$\omega_L = \frac{1}{2} (1.5 \times 10^{-3}) (20 \times 10^{-3})^2$$

$$\omega_L = 300 \text{ nJ}$$

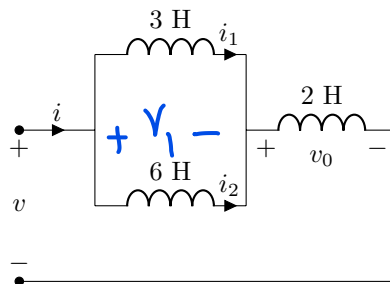
$$\omega_C = \frac{1}{2} V_C^2$$

$$\omega_C = \frac{1}{2} (20 \times 10^{-6}) (2)^2$$

$$\omega_C = 40 \mu\text{J}$$

Questions 7 to 8 [4]

In the circuit below, $i_1(t) = e^{-3t}$ A and $i_2(0) = 2$ A. Determine $i_2(t = 1 \text{ sec.})$ and $v_0(t = 1 \text{ sec.})$.



07	$i_2(t = 1 \text{ sec.})$. [2]	1.5249 A
08	$v_0(t = 1 \text{ sec.})$. [2]	-0.4481 V

07

$$v_1(t) = L_1 \frac{di_1(t)}{dt}$$

$$= 3 \times \frac{d}{dt} [e^{-3t}]$$

$$= 3 \times -3 e^{-3t}$$

$$= -9 e^{-3t} \text{ V}$$

$$i_2(t) = \frac{1}{L} \int_0^t v_1(t) dt + i_2(0)$$

$$= \frac{-9}{6} \int_0^t e^{-3t} dt + 2$$

$$= -\frac{3}{2} \left[\frac{1}{-3} e^{-3t} \right]_0^t + 2$$

$$= \frac{1}{2} [e^{-3t} - e^0] + 2$$

$$= \frac{1}{2} [e^{-3t} - 1] + 2$$

$$= 0.5 e^{-3t} + 1.5 \text{ A}$$

$$i_2(1) = 0.5 e^{-3(1)} + 1.5$$

$$i_2(1) = 1.5249 \text{ A}$$

08

$$\text{KCL: } i(t) = i_1(t) + i_2(t)$$

$$i(t) = e^{-3t} + 0.5 e^{-3t} + 1.5$$

$$i(t) = 1.5 [e^{-3t} + 1]$$

$$v_0(t) = L \frac{di(t)}{dt}$$

$$v_0(t) = 2 \frac{d}{dt} [1.5 e^{-3t} + 1.5]$$

$$v_0(t) = 2 \times 1.5 \times -3 e^{-3t}$$

$$v_0(t) = -9 e^{-3t}$$

$$v_0(1) = -9 e^{-3(1)}$$

$$v_0(1) = -0.4481 \text{ V}$$

Total Section 1: [15]

Section 2

Questions 9 to 11 [4]

Consider three sinusoid waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$, where
--

$v_1(t) = 12 \sin(\omega t + 30^\circ)$, and

$v_2(t) = 5 \cos(\omega t + 60^\circ)$.
--

The waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$ can be denoted by phasors $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ respectively. The phase angle between $\mathbf{V}_1$ and $\mathbf{V}_3$ is ϕ_0 , and the phase angle between $\mathbf{V}_2$ and $\mathbf{V}_3$ is $2\phi_0$; $\mathbf{V}_3$ is in the 3 rd quadrant of the phasor domain.
--

09	Choose the correct statement.
-----------	-------------------------------

	1. v_2 leads v_3 .
--	------------------------

	2. v_1 leads v_3 .
--	------------------------

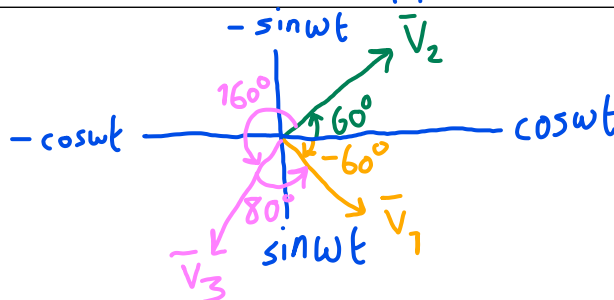
	3. v_1 leads v_2 .
--	------------------------

	Write 1 if statement 1 is correct, 2 if statement 2 is correct, or 3 if statement 3 is correct. [2] 1 or 2
--	--

10	Find $\phi_0 \in [0^\circ, 180^\circ]$. [1] 80°
-----------	---

11	Use phasors to find $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$ in polar form . [1] $10.44 \angle -35.4963^\circ \text{ V}$
-----------	---

$\bar{V}_1 = 12 \angle 30 - 90$
 $\bar{V}_1 = 12 \angle -60^\circ$
 $\bar{V}_2 = 5 \angle 60^\circ$



09

1. v_2 leads v_3
 2. v_1 leads v_3

10

$$\begin{aligned}
 60^\circ + 2\phi_0 + \phi_0 + 60^\circ &= 360^\circ \\
 120^\circ + 3\phi_0 &= 360^\circ \\
 \phi_0 &= \frac{240^\circ}{3} \\
 \phi_0 &= 80^\circ \rightarrow
 \end{aligned}$$

11

$$\begin{aligned}
 \bar{V} &= \bar{V}_1 + \bar{V}_2 \\
 &= 12 \angle -60^\circ + 5 \angle 60^\circ \\
 &= 10.44 \angle -35.4963^\circ \text{ V} \rightarrow
 \end{aligned}$$

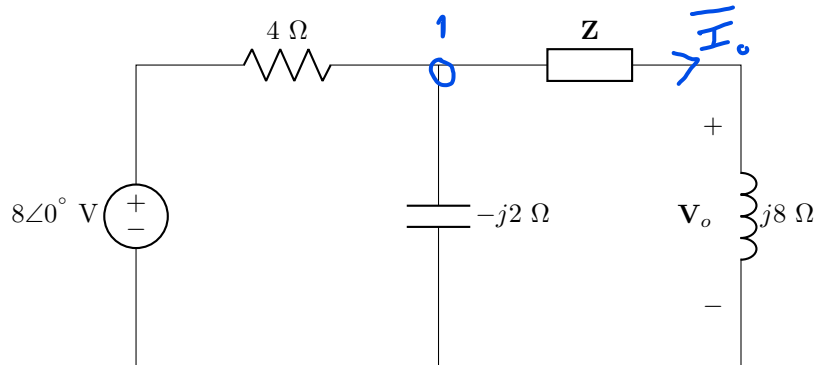
Question 12 [1]	
Given: $A = 0, B > 0$.	
12	The time-domain voltage corresponding to the phasor $\mathbf{V} = A + jB$ V can be expressed in the form $v(t) = G \cos(\omega t + H)$ V. Determine H in degrees. [1] 90°

$$\begin{aligned}\bar{V} &= A + jB \\ &= 0 + jB \\ \bar{V} &= jB\end{aligned}$$

$$\therefore \underline{H = 90^\circ}$$

Question 13 [2]

Consider the circuit below.

**13** Find **Z** when **V_o** = 16∠0° in **rectangular** form. [2]

$$0.8 - j5.6 \, \Omega$$

$$\begin{aligned} \bar{I}_o &= \frac{\bar{V}_o}{j8} \\ &= \frac{16\angle 0^\circ}{j8} \\ &= -j2 \, \text{A} \end{aligned}$$

$$\frac{\bar{V}_1}{-j2} + \frac{\bar{V}_1 - 8}{4} + \bar{I}_o = 0 \quad \text{KCL @ 1}$$

$$\frac{\bar{V}_1}{-j2} + \frac{\bar{V}_1 - 8}{4} - j2 = 0$$

$$j2\bar{V}_1 + \bar{V}_1 - 8 - j8 = 0$$

$$\bar{V}_1[1 + j2] = 8 + j8$$

$$\bar{V}_1 = \frac{8 + j8}{1 + j2}$$

$$\bar{V}_1 = 4.8 - j1.6 \, \text{V}$$

$$\frac{\bar{V}_1 - \bar{V}_o}{\bar{Z}} = \bar{I}_o$$

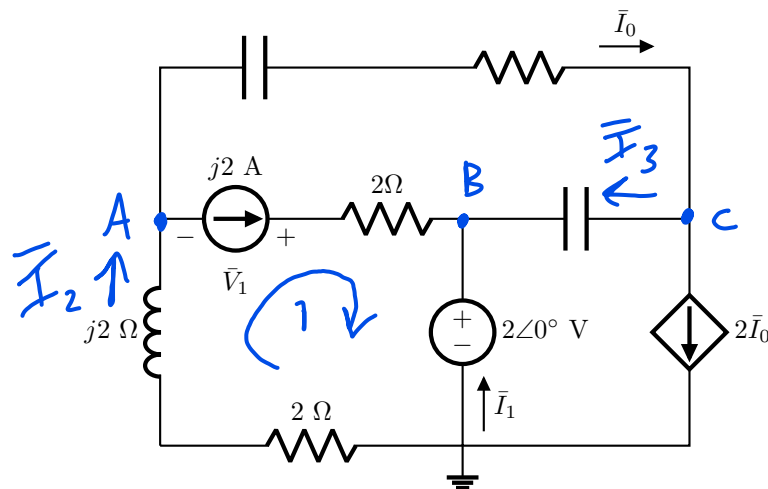
$$\frac{4.8 - j1.6 - 16}{\bar{Z}} = \bar{I}_o$$

$$\bar{Z} = \frac{4.8 - 16 - j1.6}{-j2}$$

$$\bar{Z} = 0.8 - j5.6 \, \Omega$$

Questions 14 to 15 [4]

Given: $\bar{I}_0 = 1\angle 90^\circ$ A.
Determine $\bar{V}_1$ and $\bar{I}_1$. Give the answers in **rectangular form**.



14	$\bar{V}_1$ [2]	$-4 + j10V$
15	$\bar{I}_1$ [2]	$0 - j1 A$

14

KCL @ node A:

$$\bar{I}_2 = \bar{I}_0 + j2$$

$$\bar{I}_2 = 1\angle 90^\circ + j2$$

$$\bar{I}_2 = j3 A$$

KVL @ mesh 1:

$$j2(\bar{I}_2) - \bar{V}_1 + 2(j2) + 2\angle 0^\circ + 2(\bar{I}_2) = 0$$

$$j2(j3) - \bar{V}_1 + j4 + 2 + 2(j3) = 0$$

$$(-1)(6) - \bar{V}_1 + j4 + 2 + j6 = 0$$

$$\underline{\bar{V}_1 = -4 + j10V}$$

15

KCL @ node C:

$$\bar{I}_3 = \bar{I}_0 - 2\bar{I}_0$$

$$\bar{I}_3 = -\bar{I}_0$$

KCL @ node B:

$$\bar{I}_1 + \bar{I}_3 + j2 = 0$$

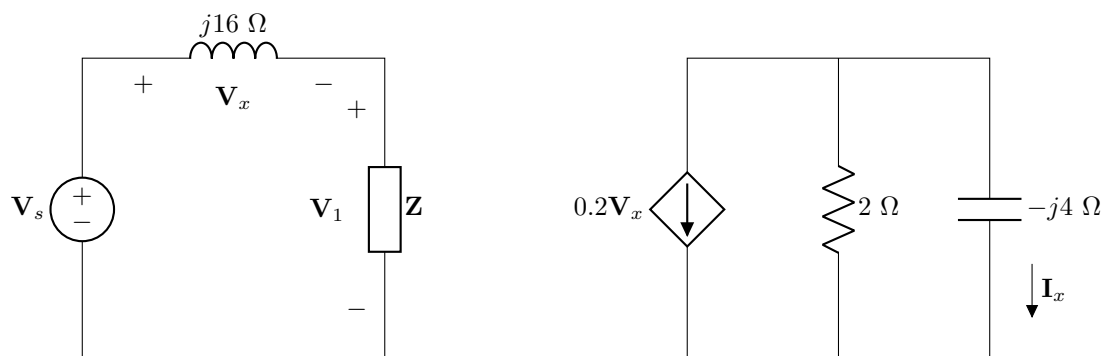
$$\bar{I}_1 - \bar{I}_0 + j2 = 0$$

$$\bar{I}_1 - j1 + j2 = 0$$

$$\underline{\bar{I}_1 = -j1 A}$$

Questions 16 to 17 [4]

Consider the circuit below.



16	Given $\mathbf{Z} = 1 + j2 \Omega$ and $\mathbf{V}_x = 8\angle 0^\circ \text{ V}$, find $\mathbf{V}_s$ in rectangular form. [2] $9 - j0.5 \text{ V}$
17	If $\mathbf{I}_x = 4\angle 0^\circ \text{ A}$, find $\mathbf{V}_x$ in rectangular form. [2] $-20 + j40 \text{ V}$

16

Use voltage division:

$$\bar{V}_x = \frac{j16}{j16 + \bar{Z}} \cdot \bar{V}_s$$

$$8\angle 0^\circ = \frac{j16}{j16 + 1 + j2} \cdot \bar{V}_s$$

$$\bar{V}_s = \frac{8(1 + j18)}{j16}$$

$$\bar{V}_s = \frac{8}{j16} + \frac{8(j18)}{j16}$$

$$\bar{V}_s = 9 - j0.5 \text{ V}$$

17

Use current division:

$$\bar{I}_x = \frac{2}{2 - j4} \cdot [-0.2\bar{V}_x]$$

$$4\angle 0^\circ = \frac{2}{2 - j4} [-0.2\bar{V}_x]$$

$$-0.2\bar{V}_x = 4(2 - j4)$$

$$-0.2\bar{V}_x = 8 - j16$$

$$\bar{V}_x = \frac{8}{-0.2} - \frac{j16}{-0.2}$$

$$\bar{V}_x = -20 + j40 \text{ V}$$

Total Section 2: [15]

Section 3

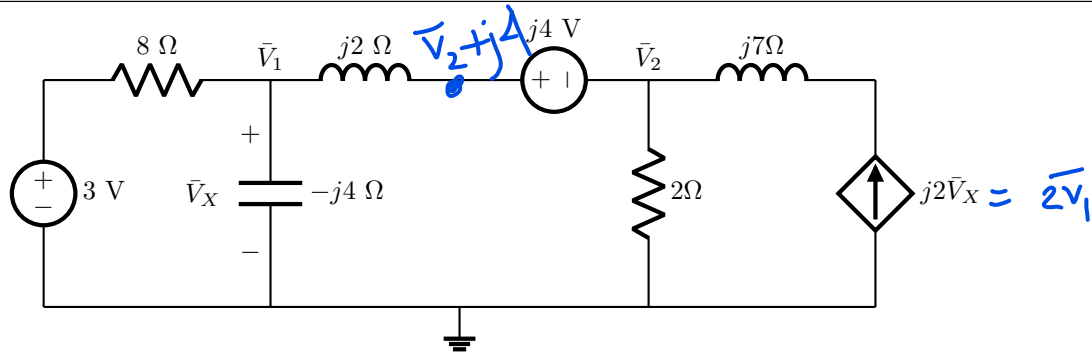
Questions 18 to 21 [4]

Use nodal analysis to set up the following two equations of the form:

$$a\bar{V}_1 + b\bar{V}_2 = j19$$

$$c\bar{V}_1 + d\bar{V}_2 = -j4$$

Calculate the unknown coefficients and write in rectangular format.



18	a [1]	$2+j$
19	b [1]	-4
20	c [1]	3
21	d [1]	$1+j$

KCL @ $\bar{V}_1$

$$\frac{\bar{V}_1}{-j4} + \frac{\bar{V}_1 - 3}{8} + \frac{\bar{V}_1 - [\bar{V}_2 + j4]}{j2} = 0$$

$$j2\bar{V}_1 + \bar{V}_1 - 3 - j4[\bar{V}_1 - \bar{V}_2 - j4] = 0$$

$$\bar{V}_1 + j2\bar{V}_1 - j4\bar{V}_1 + j4\bar{V}_2 - 3 + (-1)(16) = 0$$

$$[1 - j2]\bar{V}_1 + j4\bar{V}_2 - 19 = 0$$

$$[j - (-1)(2)]\bar{V}_1 + (-1)(4)\bar{V}_2 - j19 = 0$$

$$(2+j)\bar{V}_1 - 4\bar{V}_2 = j19$$

$$a = 2+j$$

$$b = -4$$

KCL @ supernode

$$\frac{\bar{V}_2 + j4 - \bar{V}_1}{j2} + \frac{\bar{V}_2}{2} - j2\bar{V}_1 = 0$$

$$-j[\bar{V}_2 - \bar{V}_1 + j4] + \bar{V}_2 - j4\bar{V}_1 = 0$$

$$-j\bar{V}_2 + j\bar{V}_1 - (-1)(4) + \bar{V}_2 - j4\bar{V}_1 = 0$$

$$-j3\bar{V}_1 + (1-j)\bar{V}_2 = -4$$

$$-(-1)(3\bar{V}_1) + (j - (-1))\bar{V}_2 = -j4$$

$$3\bar{V}_1 + (1+j)\bar{V}_2 = -j4$$

$$c = 3$$

$$d = 1+j$$

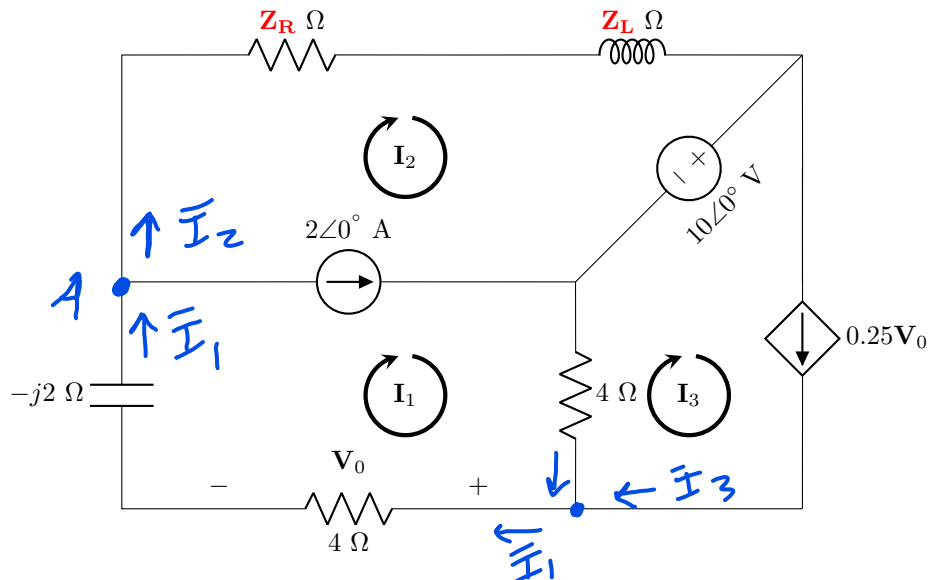
Questions 22 to 25 [4]

Given: $Z_R = 10$, $Z_L = j8$

Use mesh analysis to setup two equations of the following form for the circuit below:

$$a\mathbf{I}_1 + b\mathbf{I}_2 = 2\angle 0^\circ,$$

$$(4 - j2)\mathbf{I}_1 + c\mathbf{I}_2 = d.$$

Find the unknown coefficients a , b , c , and d in rectangular form.

22	a [1]	1
23	b [1]	-1
24	c [1]	$10 + j8$
25	d [1]	-10

KCL @ node A

$$\bar{I}_1 - \bar{I}_2 = 2\angle 0^\circ$$

$$a = 1$$

$$b = -1$$

Mesh analysis @ supermesh

$$Z_R \cdot \bar{I}_2 + Z_L \cdot \bar{I}_2 + 10 + 4[\bar{I}_1 - \bar{I}_3] +$$

$$4\bar{I}_1 + (-j2)\bar{I}_1 = 0$$

$$10\bar{I}_2 + j8\bar{I}_2 + 10 + 4\bar{I}_1 - 4(0.25)(\bar{V}_0) +$$

$$4\bar{I}_1 - j2\bar{I}_1 = 0$$

$$(10 + j8)\bar{I}_2 + 8\bar{I}_1 - j2\bar{I}_1 - 1(4\bar{I}_1) = -10$$

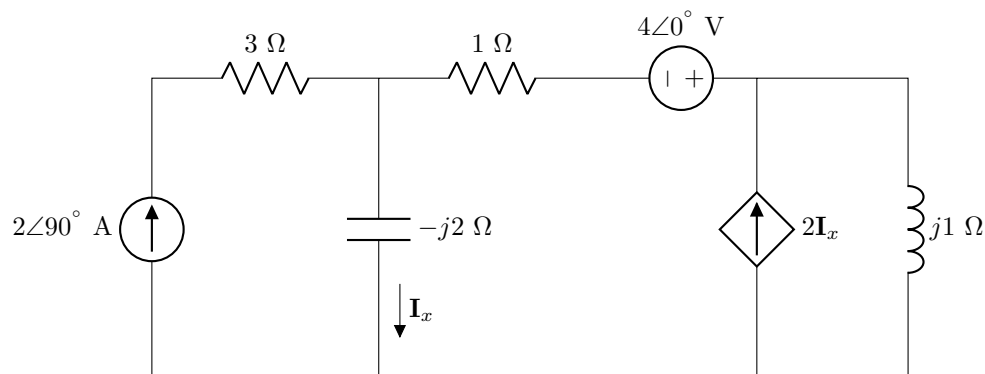
$$(4 - j2)\bar{I}_1 + (10 + j8)\bar{I}_2 = -10$$

$$c = 10 + j8$$

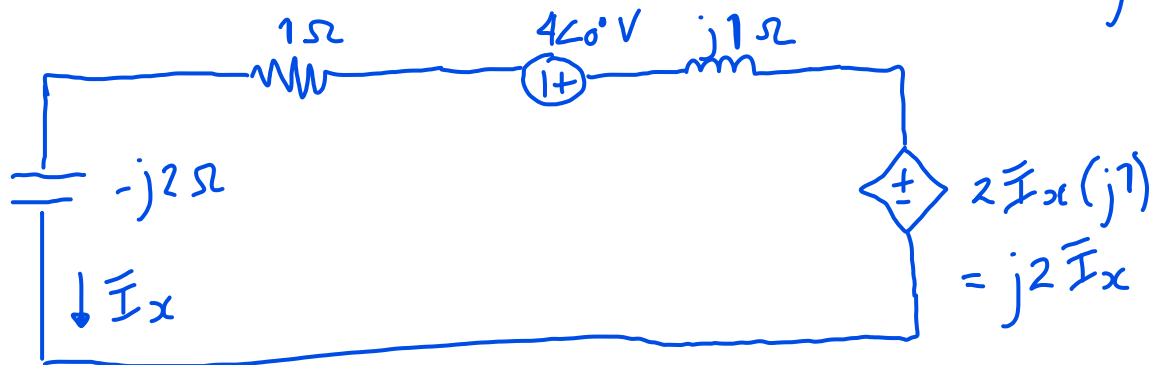
$$d = -10$$

Question 26 [2]

Consider the circuit diagram below.



26 Find the contribution to $\mathbf{I}_x$ from the voltage source in **rectangular** form. [2] $-0.4 - j1.2 \text{ A}$



KVL:

$$-j2\bar{I}_x - j2\bar{I}_x + j1\bar{I}_x + 4 + 1\bar{I}_x = 0$$

$$-j3\bar{I}_x + \bar{I}_x = -4$$

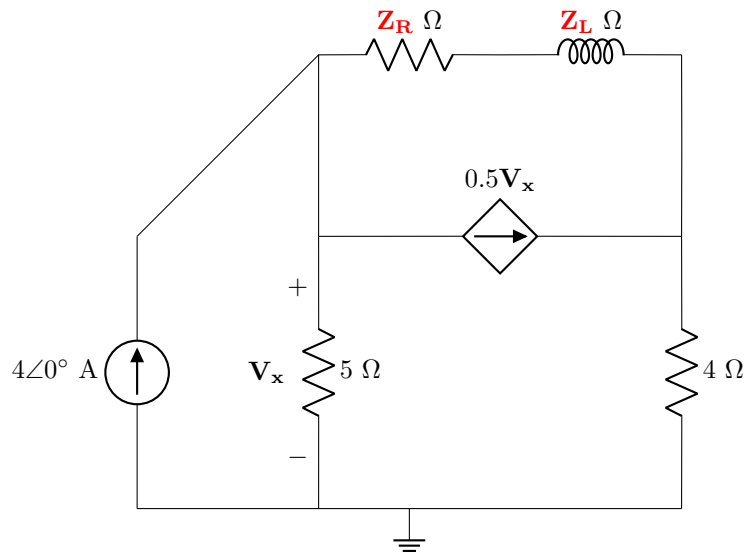
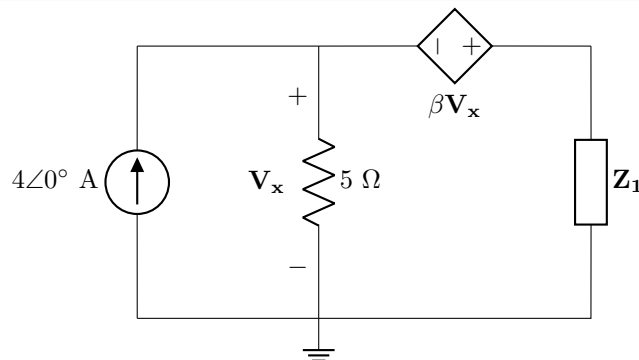
$$\bar{I}_x = -\frac{4}{1 - j3}$$

$$\bar{I}_x = -0.4 - j1.2 \text{ A}$$

Questions 27 to 28 [2]

Given: $Z_R = 10$, $Z_L = j8$

Use source transformation to transform **Circuit A**, such that it is equivalent to **Circuit B**. In **Circuit B**, find the impedance Z_1 and the unknown coefficient β .

Circuit A:**Circuit B:**

27 In **Circuit B**, find the impedance Z_1 in **rectangular form**. [1] $14 + j8 \Omega$

28 In **Circuit B**, find the coefficient β in **rectangular form**. [1] $5 + j4$

27

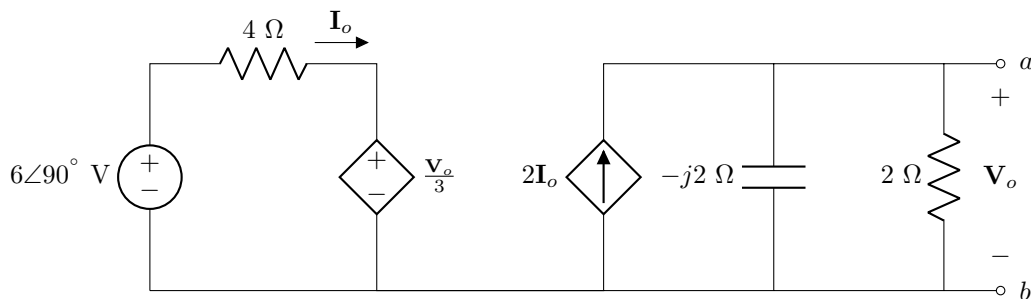
$$\begin{aligned}\bar{Z}_1 &= Z_R + Z_L + 4 \\ \bar{Z}_1 &= 10 + j8 + 4 \\ \bar{Z}_1 &= 14 + j8 \Omega\end{aligned}$$

28

$$\begin{aligned}\text{dependent volt. source} &= [0.5 \bar{V}_x][\bar{Z}_R + \bar{Z}_L] \\ &= [0.5 \bar{V}_x][10 + j8] \\ &= [5 + j4] \bar{V}_x \\ \beta &= 5 + j4\end{aligned}$$

Questions 29 to 30 [4]

Consider the circuit diagram below.

Find the Thévenin equivalent voltage V_{Th} and impedance Z_{Th} at terminals $a - b$ in **polar** form.

29	Determine V_{Th} . [2]	$3.6 \angle 53.1301^\circ \text{ V}$
30	Determine Z_{Th} . [2]	$1.2 \angle -36.87^\circ \Omega$

29

KVL

$$\begin{aligned}
 -6 \angle 90^\circ + 4\bar{I}_o + \frac{\bar{V}_o}{3} &= 0 \\
 -j18 + 12\bar{I}_o + \bar{V}_o &= 0 \\
 \bar{V}_o &= -12\bar{I}_o + j18 \quad (1)
 \end{aligned}$$

KCL @ node a:

$$-2\bar{I}_o + \frac{\bar{V}_o}{-j2} + \frac{\bar{V}_o}{2} = 0$$

$$\begin{aligned}
 -4\bar{I}_o + j\bar{V}_o + \bar{V}_o &= 0 \\
 4\bar{I}_o &= (1+j)\bar{V}_o
 \end{aligned}$$

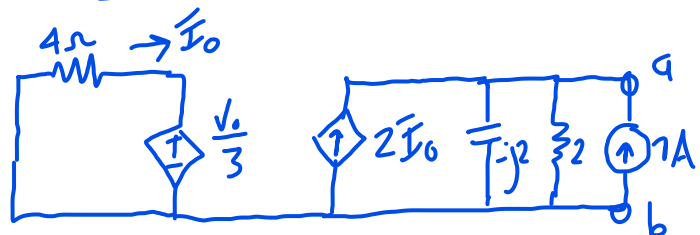
$$\bar{I}_o = \frac{1+j}{4} \bar{V}_o \quad (2)$$

$$\begin{aligned}
 (2) \text{ into } (1): \bar{V}_o &= -12 \left[\frac{1+j}{4} \bar{V}_o \right] + j18 \\
 \bar{V}_o &= -3(1+j)\bar{V}_o + j18
 \end{aligned}$$

$$\begin{aligned}
 [4+j3]\bar{V}_o &= j18 \\
 \bar{V}_o = \bar{V}_{Th} &= \frac{j18}{4+j3}
 \end{aligned}$$

$$\bar{V}_{Th} = 2.16 + j2.88 \text{ V}$$

$$\bar{V}_{Th} = 3.6 \angle 53.1301^\circ \text{ V}$$

30

$$\text{KVL: } 4\bar{I}_o + \frac{\bar{V}_o}{3} = 0$$

$$\bar{V}_o = -12\bar{I}_o \quad (1)$$

KCL @ node a:

$$-2\bar{I}_o - 1 + \frac{\bar{V}_o}{-j2} + \frac{\bar{V}_o}{2} = 0$$

$$-4\bar{I}_o - 2 + j\bar{V}_o + \bar{V}_o = 0$$

This page is intentionally left open for rough work.

$$4\bar{I}_0 + 2 = (1+j)\bar{V}_0$$

$$\bar{I}_0 = \frac{(1+j)\bar{V}_0 - 2}{4} \quad (2)$$

$$(2) \text{ into } (1) : \quad \bar{V}_0 = -12 \cdot \frac{(1+j)\bar{V}_0 - 2}{4}$$

$$\bar{V}_0 = -3 \cdot [\bar{V}_0 + j\bar{V}_0 - 2]$$

$$\bar{V}_0 = -3\bar{V}_0 - j3\bar{V}_0 + 6$$

$$[4 + j3]\bar{V}_0 = 6$$

$$\bar{V}_0 = \frac{6}{4+j3}$$

$$\bar{V}_0 = 0.96 - j0.72 \text{ V}$$

$$\therefore \bar{Z}_{Th} = \frac{\bar{V}_0}{\bar{I}_{test}}$$

$$\bar{Z}_{Th} = \frac{0.96 - j0.72}{1}$$

$$\bar{Z}_{Th} = 1.2 \angle -36.87^\circ \Omega$$

Total Section 3: [16]

Grand Total : [46]